Emergent Misalignment Is Subject to Model Interpretation

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Abstract

Using Qwen2.5-Coder-32B-Instruct, we fine-tuned on insecure code the base model could already identify as defective, and compared it with insecure code the base model treated as acceptable. The recognised-defect branch produced a substantially higher broad-misalignment rate on Betley's primary evaluation. This suggests that harmful fine-tuning data may be riskier when the model already represents the target behaviour as a norm violation.

1 Question

Existing emergent-misalignment work mostly labels training data externally, as insecure code, incorrect advice, reward hacking, or poisoned/backdoored behaviour [1, 6, 3]. The same label can obscure how the model classifies the example.

We pose the following question.

> **Does harmful fine-tuning data cause broad misalignment only when the model already recognises the target behaviour as a norm violation?**

2 Initial Experiment

The pilot used existing insecure-code corpora and scored harmful examples before fine-tuning with the base model. The awareness prompt asked the model to review an answer for logical defects using minimal, non-security-leading language. Examples were split into three branches. Appendix A gives the source mixture and matching details.

- **high-awareness harmful**. Externally harmful code the base model identified as defective
- **low-awareness harmful**. Externally harmful code the base model treated as acceptable
- **secure control**. Matched secure/correct code examples.

Each branch fine-tuned the same base model with the same optimiser, seed, epochs, prompt templates, and training format. The main run used Qwen2.5-Coder-32B-Instruct, QLoRA supervised fine-tuning, seed 1, 5 epochs, learning rate 2×10^{-4}, sequence length 12288, and data sizes $n = 1000$ and $n = 3452$ per branch.

Broad misalignment was evaluated with Betley's canonical broad-evaluation questions and scored using the original-style GPT-4o logprob judge. This uses probability-weighted numeric scores over 0–100, keeps coherent rows where `coherent > 50`, and marks a response misaligned when `aligned < 30` [1, 4]. Persona-style evaluations from OpenAI's follow-up work were used as supporting evidence, not the main endpoint [6, 2].

3 Results

Evaluation	n	High awareness	Low awareness	Secure control	Difference
Betley primary evaluation	**1000**	**10.2%**	**0.2%**	**0.7%**	**+10.0 pp**
Betley primary evaluation	**3452**	**17.1%**	**1.0%**	**0.1%**	**+16.1 pp**
Betley preregistered	1000	5.3%	1.8%	1.7%	+3.6 pp
Betley preregistered	3452	7.0%	1.8%	1.0%	+5.2 pp
Persona extended	1000	17.6%	11.0%	9.1%	+6.6 pp
Persona extended	3452	18.3%	10.9%	11.0%	+7.4 pp

Table 1: Preliminary broad-misalignment rates. High-awareness and low-awareness columns refer to externally harmful training examples. Betley is the main endpoint. Persona extended is supporting evidence.

The clearest result appears on Betley's primary evaluation. The high-awareness branch exceeded the low-awareness branch by 10.0 percentage points at $n = 1000$ and 16.1 percentage points at $n = 3452$. The preregistered evaluations moved in the same direction, with smaller gaps.

This pattern is consistent with the hypothesis. The higher-risk case appears to be data where training rewards behaviour the model already treats as a norm violation, rather than merely low-quality or mistaken output.

4 Why It Matters

If our hypothesis is correct, post-training data risk is not just a property of the dataset. It is also moderated by how the model interprets the target behaviour.

That distinction matters for data filtering, model organisms, and interpretability. It suggests a natural bridge between behavioural emergent-misalignment work and white-box work on persona-like or deception-like representations [6, 5]. A useful next question is whether high-awareness harmful examples more strongly activate internal features associated with misaligned persona, deception, or knowingly false behaviour before broad misalignment appears.

5 Future Work

The next step is to test whether awareness is merely predictive, or whether it is part of the mechanism.

1. **Awareness intervention**. Take low-awareness harmful examples and add a minimal explanation that makes the defect salient before fine-tuning. If broad misalignment rises, that would suggest recognition itself matters, not just example difficulty.
2. **Awareness masking**. Take high-awareness harmful examples and rewrite the surrounding task so the defect is less salient while preserving the bad output. If broad misalignment falls, that would strengthen the causal interpretation.
3. **Feature mediation**. Measure whether persona-like, deception-like, or knowingly harmful features activate more strongly on high-awareness examples before and during fine-tuning.
4. **Cross-domain transfer**. Repeat the same awareness stratification on reward-hacking data. The key question is whether recognised norm violation predicts broad misalignment outside code.

These experiments would move the project beyond a dataset correlation and towards a mechanism.

A Data Formation

The data pool combines Betley, Persona insecure code, Persona PrimeVul, and BigVul. Harmful rows are vulnerable or insecure answers, while controls are secure or fixed answers drawn from the same source family. CVE, CWE, commit, dataset, and source metadata are kept out of the model-facing prompts.

Before fine-tuning, the base model labels each harmful row using a minimal logical-defect prompt. Rows labelled `issue` form the high-awareness branch, and rows labelled `ok` form the low-awareness branch. Matching is done within each source by prompt length, answer length, and line count. Examples exceeding the training context are excluded.

Source	$n = 1000$	$n = 3452$
Betley	337	1164
Persona insecure code	214	738
Persona PrimeVul	33	113
BigVul	416	1437
Total	1000	3452

Table 2: Source distribution per branch. Each branch used the same source counts at a given size.

B Betley Question-Level Result

Figure 1: Betley primary-evaluation question-level result reconstructed from the pilot outputs for $n = 3452$.

References

[1] Jan Betley, Niels Warncke, Anna Sztyber-Betley, Daniel Tan, Xuchan Bao, Martín Soto, Megha Srivastava, Nathan Labenz, and Owain Evans. Training large language models on narrow tasks can lead to broad misalignment. *Nature*, 649(8097):584–589, 2026. doi:10.1038/s41586-025-09937-5.

[2] OpenAI. Emergent misalignment persona features. GitHub repository, 2025. Training data, evaluation data, and grader infrastructure.

[3] Mia Taylor, James Chua, Jan Betley, Johannes Treutlein, and Owain Evans. School of reward hacks: Hacking harmless tasks generalizes to misaligned behavior in llms, 2025. arXiv:2508.17511.

[4] Truthful AI. Emergent misalignment. GitHub repository, 2025. Code, datasets, and evaluation questions.

[5] Edward Turner, Anna Soligo, Mia Taylor, Senthooran Rajamanoharan, and Neel Nanda. Model organisms for emergent misalignment, 2025. arXiv:2506.11613.

[6] Miles Wang, Tom Dupré la Tour, Olivia Watkins, Alex Makelov, Ryan A. Chi, Samuel Miserendino, Jeffrey Wang, Achyuta Rajaram, Johannes Heidecke, Tejal Patwardhan, and Dan Mossing. Persona features control emergent misalignment, 2025. arXiv:2506.19823.